

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a white coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

ALLERGIC RHINITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACE Angiotensin-Converting Enzyme

ASA Acetylsalicylic Acid

CNS Central Nervous System

GI Gastrointestinal

HCP Healthcare Professional/Provider

IgE Immunoglobulin E

MAOIs Monoamine Oxidase Inhibitors

NSAIDs Nonsteroidal Anti-Inflammatory Drugs

OTC Over-the-Counter

PDE5 Phosphodiesterase Type 5

PO By Mouth

PRN As needed

SOB Shortness of Breath

WHAT IS ALLERGIC RHINITIS?

Allergic rhinitis affects patients either:

- **Seasonally**
 - Occurs during specific seasons (generally spring through fall)
 - Allergens include: pollen from plants, mold spores
- **Perennially**
 - Occurs year-round
 - Non-seasonal allergens: dust mites, animal dander, cockroaches, mold spores



Note: many patients experience symptoms of perennial allergic rhinitis that are exacerbated seasonally



Also known as **hay fever**, is the inflammation of the nasal mucosa because of environmental allergens

CLASSIFICATION

Duration of Symptoms

- **Intermittent**
 - <4 days/week OR
 - <4 consecutive weeks
- **Persistent**
 - ≥4 days/week AND
 - ≥4 consecutive weeks

Severity of Symptoms

- **Mild**
 - Symptoms present but are not bothersome and do not impair daily activities or quality of life
- **Moderate-severe**
 - Impaired school/work performance
 - Impaired daily activities
 - Sleep disturbance
 - Troublesome symptoms

Time of Year

- **Seasonal**
 - During a certain season (especially spring-fall)
- **Perennial**
 - Year-round symptoms

ALLERGIC CASCADE OF ALLERGIC RHINITIS

Sensitization

- Allergen enters the nose
- Lymphocytes detect the allergens and produce IgE antibodies
- IgE antibodies bind to mast cells

Immediate-phase reaction

- Occurs within minutes upon re-exposure of the allergen
- The allergen interacts with the IgE bound to the mast cells
- Triggers mast cells to release histamine and other inflammatory mediators

Late-phase reaction

- 4-8 hours after the exposure
- Migration of inflammatory and immune cells (eosinophils, macrophages, basophils)

CLINICAL PRESENTATION

Patients with allergic rhinitis may present with:

- Congestion
- Runny nose
- Postnasal drip
- Sneezing
- Pruritic eyes, ears, or nose

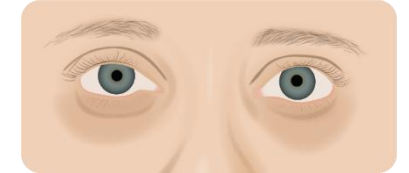
Untreated or failed treatment may result in a lower quality of life

- Sleep disturbance
- Fatigue
- Malaise
- Emotional distress: anxiety, depression, low self-esteem
- Impaired cognitive function: poor concentration, reduced work/school performance
- Impaired daily activities

CLINICAL PRESENTATION

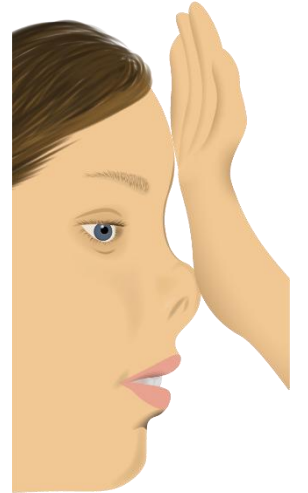
Allergic shiners

Darkened and swollen areas under the eyes due to infraorbital edema



Allergic facies

Arched palate and dental malocclusion



Allergic salute

Pushing the tip of the nose upward with the hand



Allergic crease/ transverse nasal crease

Crease caused by allergic salute

RISK FACTORS

Family history/genetic predisposition of atopy

- Allergen enters the nose

Allergen exposure

Industrialized countries

- Increased risk of developing allergies and asthma when immigrants go from developing countries to western industrialized countries

Allergic rhinitis often co-exists with the following:

- Asthma
- Allergic conjunctivitis
- Acute or chronic rhinosinusitis

DIAGNOSIS

- Most cases of allergic rhinitis can be diagnosed through history and symptoms alone
- If further evaluation is needed, then the following tests can be used for diagnosis:
 - Physical exam: physician examines the nose, oropharynx, tympanic membrane, and eyes
 - Allergen-specific testing: skin tests, serum tests

MEDICAL AND DRUG RELATED CAUSES

Non-Allergic Rhinitis

- **Autonomic**
 - Drug-induced
 - Hormonal
 - Vasomotor
- **Infectious**
 - Viral
 - Bacterial
 - Fungal
- **Idiopathic**

Drug-Induced Rhinitis

- **Autonomic**
 - Topical Decongestants (rhinitis medicamentosa)
 - Alpha blockers
- **ACE inhibitors**
- **Gabapentin**
- **Beta blockers**
- **Chlorpromazine**
- **PDE-5 inhibitors**
- **ASA/NSAIDs**
- **Hormonal contraceptives**

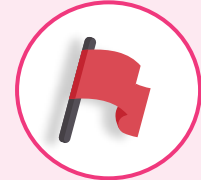
GOALS OF THERAPY

- ✓ Prevent allergic reaction and rhinitis by avoiding exposure to allergens
- ✓ Suppress and control symptoms produced by allergic response
- ✓ Avoid/minimize side effects of medication including impact on activities of daily living

RED FLAGS

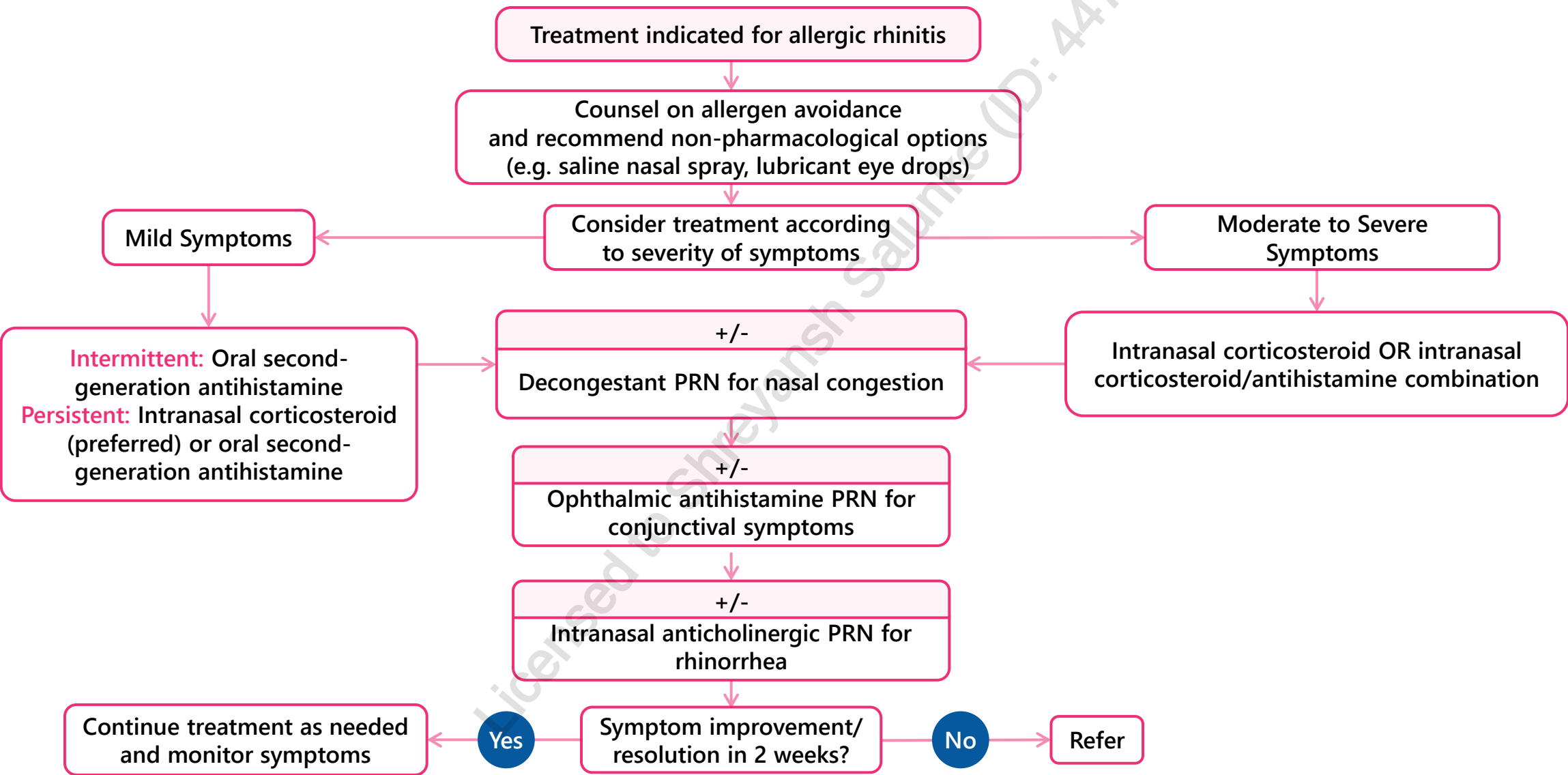
Refer IF:

- Children <2 years old
- Treatment failure/persistent symptoms
 - Try non-prescription medication for 2 weeks
- Severe allergic rhinitis affecting quality of life
- Unilateral symptoms
- Concomitant medical condition (e.g. otitis media, asthma, recurrent sinusitis)
- Unable to identify allergen
- Fever
- Facial pain
- Loss of taste or smell



Caution in advising certain self-care products during pregnancy due to possible teratogenic effects

TREATMENT ALGORITHM



NON-PHARMACOLOGICAL MEASURES

Avoid allergens whenever possible

- Outside allergens: remain indoors as much as possible, air conditioning, keep windows and doors closed, do not dry clothes outside
- Indoor allergens: avoid carpeting and furnished areas, vacuum with a facemask or have someone else vacuum, wash bedding in hot cycle every 2 weeks

Avoid aggravating factors such as smoke, air pollution, and insect sprays

Nasal saline irrigation, lubricant eye drops (conjunctival symptoms)

PHARMACOLOGICAL MEASURES - OTC PRODUCTS

Oral antihistamines

- Used to treat rhinorrhea, sneezing, itching, conjunctivitis
- Not used as monotherapy for nasal congestion

Decongestants

- Used to treat congestion

Intranasal corticosteroids

- Triamcinolone acetonide and fluticasone propionate nasal sprays (Nasacort AQ® and Flonase®) approved in Canada as OTC options

PHARMACOLOGICAL MEASURES - OTC PRODUCTS

ORAL ANTIHISTAMINES

Drugs: first-generation antihistamines	MOA	Comments
• Diphenhydramine (Benadryl®) • Chlorpheniramine • Cyproheptadine • Brompheniramine • Only available as combination products	Competitive antagonists for H1 receptor, blocks histamine	• Not recommended first-line due to adverse effects • Side effects: CNS (sedation, fatigue, impaired cognitive function), anticholinergic (urinary retention, dry eyes, dry mouth, constipation)

PHARMACOLOGICAL MEASURES - OTC PRODUCTS

ORAL ANTIHISTAMINES

Drugs: second-generation antihistamines	MOA	Comments
• Cetirizine (Reactine®) • Loratadine (Claritin®) • Desloratadine (Aerius®) • Fexofenadine (Allegra®) • Bilastine (Blexten®) - RX • Rupatadine (Rupall®) -RX	Competitive antagonists for H1 receptor, blocks histamine	• First-line therapy for mild intermittent allergic rhinitis • May need additional therapy for nasal congestion • Onset approximately 1 hour • Most effective if taken prophylactically (1-2 hours before exposure) or at start of exposure • Reduces rhinorrhea, sneezing, and itching • Caution in renal impairment, may require dose adjustment • Side effects: headaches, less CNS and anticholinergic effects compared to first-generation antihistamines, however, some sedation is possible with high-dose cetirizine

PHARMACOLOGICAL MEASURES - OTC PRODUCTS

ORAL DECONGESTANTS

Drugs	MOA	Comments
• Phenylephrine • Pseudoephedrine	Adrenergic agonist causes alpha-1 mediated vasoconstriction of the nasal mucosa	• Most commonly in combination products with antihistamines or analgesics • Not as fast or effective as topical decongestants • Avoid use in patients with uncontrolled hypertension or hyperthyroidism and pregnancy • Do not use within 2 weeks of taking MAOIs • Side effects: restlessness, dizziness, insomnia, tachycardia, increased blood pressure (caution in patients with hypertension), dysglycemia (caution in patients with diabetes)

PHARMACOLOGICAL MEASURES - OTC PRODUCTS

TOPICAL DECONGESTANTS

Drugs	MOA	Comments
• Phenylephrine (Dristan®) • Short-acting (up to 4 hours) • Oxymetazoline (Drixoral®) • Long-acting agent (up to 12 hours) • Xylometazoline (Otrivin®) • Long-acting agent (up to 12 hours)	Adrenergic agonist causes alpha-1 mediated vasoconstriction of the nasal mucosa	• Fast-acting and effective • Onset 5-10 minutes • Do not use for more than 3-5 days due to rebound congestion • Side effects: nasal irritation (sneezing, dryness, burning, stinging)

PHARMACOLOGICAL MEASURES - RX PRODUCTS

Intranasal corticosteroids

Intranasal anticholinergics

Intranasal antihistamines +corticosteroids

Leukotriene receptor antagonists

Immunotherapy

PHARMACOLOGICAL MEASURES - RX PRODUCTS

INTRANASAL CORTICOSTEROIDS

Drugs	MOA	Comments
• Budesonide (Rhinocort®) • Beclomethasone • Ciclesonide (Omnaris®) • Fluticasone propionate (Flonase®) • Fluticasone furoate (Avamys®) • Mometasone (Nasonex®) • Triamcinolone (Nasacort AQ®)	Downregulates inflammatory response to inhibit inflammation	• First-line therapy for mild-persistent or moderate-severe allergic rhinitis • Patients may feel improvement in a few days, but peak effect may take 1-2 weeks, therefore counsel patients to take prior to exposure if possible • Improves nasal congestion, rhinorrhea, ocular symptoms and lower airway symptoms (in patients with asthma) • Anosmia is not usually improved • Limited systemic absorption • Side effects: nasal irritation, stinging, nose bleeds. Pointing nozzle away from nasal septum can reduce irritation. • Can use with oral antihistamine

PHARMACOLOGICAL MEASURES - RX PRODUCTS

INTRANASAL ANTICHOLINERGICS

Drugs	MOA	Comments
Ipratropium bromide	Inhibits secretion by blocking acetylcholine	- Effective treatment for rhinorrhea - Onset: within 15 minutes - Usually best suited for vasomotor rhinitis - Side effects: anticholinergic effects (e.g. dry mouth/throat, nasal dryness, nasal irritation, nose bleeds, headaches)

PHARMACOLOGICAL MEASURES - RX PRODUCTS

INTRANASAL ANTIHISTAMINES COMBINED WITH INTRANASAL CORTICOSTEROIDS

Drugs	MOA	Comments
- Azelastine Hydrochloride and Fluticasone Propionate (Dymista®) - Olopatadine Hydrochloride and Mometasone Furoate Nasal Spray (Ryaltris®)	Competitive antagonists for H1 receptor, blocks histamine. Also downregulates inflammatory response to inhibit inflammation.	- Usually used if intranasal corticosteroid alone did not work - Twice daily dosing - Onset: within 15 minutes - Side effects: nasal irritation (burning, dryness, pain), nosebleeds, headaches, sinusitis

PHARMACOLOGICAL MEASURES - RX PRODUCTS

LEUKOTRIENE RECEPTOR ANTAGONISTS

Drugs	MOA	Comments
Montelukast (Singulair®)	Inhibits cysteinyl leukotrienes- LTC ₄ , LTD ₄ , LTE ₄ (inflammatory cells)	- Once daily dosing, available as chewable - Helpful in management of allergic rhinitis concomitant with asthma or nasal polyps - Symptom relief is greater when added with second generation antihistamine than either medication alone - Recommended therapy if other treatments are intolerable or fail. Less effective than intranasal corticosteroids. - Side effects: headaches, GI effects (e.g., abdominal pain, diarrhea, dyspepsia, nausea, vomiting), psychiatric symptoms (rare; agitation, aggression, depression)

PHARMACOLOGICAL MEASURES - RX PRODUCTS

OPHTHALMIC PREPARATIONS

Drugs	Comments
• Topical ophthalmic antihistamines: olopatadine, ketotifen • Onset: few minutes • Duration of action: 12 hours • Topical mast cell stabilizers: cromoglycate, lodoxamide • Onset: several days • QID administration	• May be used when PO antihistamines, intranasal corticosteroids, and leukotriene receptor antagonists do not improve conjunctival symptoms • Cromoglycate should be discarded within a month after opening

PHARMACOLOGICAL MEASURES - RX PRODUCTS

IMMUNOTHERAPY

- Injections of increasing doses of antigens to induce tolerance of the allergen
- Indicated for patients whose allergic rhinitis is severe, difficult to control with treatment, and cannot avoid the allergen
- Costly and requires a significant amount of time
- Risk of anaphylactic reactions during treatment

SPECIAL POPULATIONS - PREGNANCY

Antihistamines

- Oral antihistamines appear safe for use in pregnancy (e.g. cetirizine, loratadine, chlorpheniramine)

Corticosteroids

- **Intranasal** corticosteroids are considered safe during pregnancy
- Potential risk associated with triamcinolone in 1st trimester

Leukotriene receptor antagonists

- Considered generally safe during pregnancy

Decongestants

- **Intranasal** preferred over oral
- Avoid oral decongestant use in 1st trimester

Allergic rhinitis can worsen in pregnancy



COUNSELLING TIPS - NASAL SPRAY

1. Gently blow your nose to remove any mucous
2. Shake the bottle, remove the cap, and prime if necessary
3. Have your head in the upright position
4. Press down on one nostril using your finger to close the nostril
5. Place the tip of the spray into the open nostril facing away from the septum
6. Squeeze the bottle and breathe in slowly through your nose
7. Remove the tip and repeat on the other side
8. Rinse the tip with hot water and dry it before putting the cap back



MONITORING PARAMETERS

EFFICACY ENDPOINTS

Parameter	Degree of Change	Timeframe
Nasal symptoms (e.g. sneezing, rhinorrhea, itching)	Eliminate/improve	1-2 weeks → Patient to monitor daily
Daily activities, sleep	Return to normal performance	→ HCP to monitor after 1 week

SAFETY ENDPOINTS

Parameter	Degree of Change	Timeframe
Anticholinergic effects, sedation, drowsiness, headache, insomnia	Minimal to none	Duration of therapy
Blood pressure	No elevation from baseline	Patient to monitor daily HCP to follow up with patient
Rebound congestion (with topical decongestants)	None	3-5 days

TAKEAWAY

- First-line therapy for mild persistent or moderate to severe allergic rhinitis: intranasal corticosteroids
- If intranasal corticosteroids are ineffective, could try combination product with intranasal antihistamine (if not already taking oral antihistamine)
- Do not use topical decongestants for more than 3-5 days (prevent rebound congestion)
- Do not use oral decongestants during pregnancy, lactation, or in patients with uncontrolled hypertension or hyperthyroidism
- Add decongestant if oral antihistamine does not sufficiently relieve congestion
- Use antihistamines and intranasal corticosteroids prophylactically

CASE

TU is a 32-year-old female who presents to your clinic in mid-April with a five-week history of nasal congestion, frequent sneezing, itchy eyes, and post-nasal drip. She reports that her symptoms are most pronounced in the morning and when she is outdoors, particularly while gardening or jogging, and she experiences these symptoms on 6 or 7 days per week. She also experiences occasional nighttime coughing and difficulty sleeping, which she attributes to nasal congestion.

TU works as a Grade 2 teacher and lives with her husband and two indoor cats, which she has had for the past three years. Her medical history is significant for type 2 diabetes mellitus, diagnosed four years ago, for which she takes metformin 500 mg twice daily. She is also on combined oral contraceptives (ethinyl estradiol 35 mcg/norethindrone 1 mg) for birth control. For her allergy symptoms, she has been using loratadine 10 mg daily and saline nasal spray occasionally but reports only mild relief. She denies any known drug allergies.

1. Based on TU's presentation, which of the following best describes her allergic rhinitis?
 - a) Mild intermittent
 - b) Moderate-severe intermittent
 - c) Mild persistent
 - d) Moderate-severe persistent



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2. Which of the following would be the MOST appropriate next step in managing TU's allergic rhinitis?
 - a) Switch to cetirizine 10 mg daily
 - b) Add olopatadine eye drops
 - c) Start mometasone nasal spray once daily
 - d) Use xylometazoline nasal spray twice daily



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3. If TU were pregnant and presenting with the same allergic rhinitis symptoms, which of the following would be the MOST appropriate recommendation?
- a) Triamcinolone nasal spray
 - b) Budesonide nasal spray
 - c) Pseudoephedrine orally
 - d) Montelukast orally



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CHANGE LOG

June 2025

- Formatting and grammar changes throughout
- Slide 3: Added a legend
- Slide 8: Removed “gothic rash”
- Slide 13: Updated red flags
- Slide 14: Updated treatment algorithm
- Slide 18: Added rupatadine
- Slide 22: Removed “flunisolide” product not available in Canada
- Slide 23: Removed “Atrovent” product not available in Canada
- Slide 26: Removed “nedocromil” product not available in Canada
- Slide 28: Updated pregnancy recommendations
- Slide 30: Updated takeaway points
- Slides 31, 32: Added case and questions
- Slide 33: Updated references

September/October 2025

- Slide 3: Updated legend
- Slide 5: Updated definition of persistent allergic rhinitis
- Slide 30: Added monitoring parameters
- Slides 32, 33, 34: Added days per week symptoms experienced to the case

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 5: Minor updates to definitions of persistent symptoms (changed sign to $\geq$) and mild symptoms
- Slide 14: Updated algorithm